
Response Geometry in Integrated Gradients: A Diffusion-Policy Case Study

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Abstract

Attribution evaluation depends on both the ranking that selects interventions and the response that scores them. We analyze this distinction for Integrated Gradients (IG) of a fixed-noise diffusion policy’s predicted action chunk. For quadratic and stabilized negative- ℓ_2 action-discrepancy targets, we derive an exact normalized-response relation and the positive, path-dependent factor relating their gradients. A smooth construction reverses the IG ranking despite pointwise collinearity of the nonzero gradients. A finite-grid counterexample also rules out a universal random-ranking AUC reference of 0.5. Reanalysis of 99 preserved Robotics Diffusion Transformer files shows the practical effect: changing only the response changes median vision deletion AUC from 0.4479 to 0.2908 for the same recorded quadratic-target rankings. Across eight paired rescoring cases, every median response change is positive, but overshooting tails make six means negative. Exploratory episode-omission checks leave insertion-mean signs unchanged but can reverse language-deletion signs. A second smooth construction shows that several integration grids can agree in completeness, ranking and perturbation curves while reversing the exact IG order; the inaccurate order even scores better on both perturbation metrics in this example. Existing actual-model diagnostics likewise separate repeatability from unresolved integration-budget sensitivity in an incomplete cohort. These results establish distinct roles for response geometry, numerical accuracy and ranking quality. They do not qualify a production numerical setting, demonstrate superior attribution rankings, or explain task success or learned-weight dependence. The artifact preserves the raw-record lineage, analytic checks, adverse results and provenance limits.

1 Introduction

Vision-language-action (VLA) models such as RT-2 (Zitkovich et al., 2023), Octo (Octo Model Team et al., 2024), and the Robotics Diffusion Transformer (RDT) (Liu et al., 2025) combine sensory observations and instructions to predict robot actions.¹ Diffusion policies (Chi et al., 2023) generate action chunks through iterative denoising (Ho et al., 2020). Applying a scalar attribution method to this vector-valued output requires specifying what aspect of the action prediction is being explained. That choice also matters when evaluating the resulting explanation.

Integrated Gradients (IG) (Sundararajan et al., 2017) integrates a scalar output’s gradients from a baseline to an input. For a fixed-noise action sample, one possible scalar is its discrepancy from a detached reference action. A quadratic discrepancy and a negative norm express related notions of change, but they define different scalar functions. A perturbation evaluation introduces a second choice: the response used to score the feature removals or insertions selected by an attribution ranking. A comparison that changes both choices cannot identify a ranking improvement from its native-response AUC alone.

We examine this problem through exact derivations, constructive examples and a retrospective RDT case study. Changing the response from a quadratic discrepancy to a stabilized negative ℓ_2 discrepancy changes normalized AUC even when the selected features and perturbed outputs stay fixed. At the same time,

¹AI tools were used to develop and check mathematical constructions, design and implement saved-data analyses, review code and literature, and draft and revise this manuscript. The author is responsible for the content.

the targets’ positive, path-dependent gradient relationship permits their IG rankings to differ. These statements address distinct operations: rescoring existing interventions and recomputing attributions. Neither establishes which ranking is more useful under a common response.

A third issue is whether a computed attribution accurately approximates the specified mathematical quantity. Small scalar completeness residuals can conceal coordinate errors, and repeated finite-precision calculations can reproduce the same inaccurate approximation. A smooth construction in Appendix B strengthens this point: several nested grids can agree exactly in completeness, ranking and perturbation responses while reversing the exact IG ranking. We report already collected numerical diagnostics with their full planned roster and unfinished cells, rather than using a subset of successful checks to certify the implementation. The intended attribution is a fixed-context action-discrepancy quantity; it does not explain eventual task success or establish safe robot behavior.

Our contributions are as follows.

1. **Response and path-weight analysis.** An exact normalized-response identity, its overshoot and zero-gap qualifications, and a finite-stabilizer construction in which the quadratic and negative-norm targets reverse IG feature order.
2. **A finite-grid calibration counterexample.** A 100-feature equal-contribution policy has quadratic-response insertion and deletion AUC of 0.65976 for every ordering on the historical grid, so 0.5 is not a universal chance reference.
3. **An authenticated retrospective case study.** Within-ranking rescoring of preserved RDT perturbation curves, with explicit occurrence rules, episode grouping and result-to-record lineage. Unknown historical context identities prevent a paired comparison of ranking quality.
4. **Numerical diagnostic limits.** A smooth counterexample to certification by finite-grid agreement, together with fixed-input repeatability and integration-budget checks from a partially completed campaign. Failures and incomplete coverage remain visible; no production setting is approved.

The contribution is the concrete separation of these issues and its documented consequences in one diffusion-policy family. A common-response ranking study would answer an additional efficacy question; its absence limits that conclusion without invalidating the fixed-ranking analysis.

2 Related Work

2.1 Attribution Methods

Sundararajan et al. (2017) introduced IG with the sensitivity and implementation-invariance axioms and showed that straight-line path integration satisfies both. Later work has addressed limitations of the vanilla method. Jha et al. (2021) computed attributions over neural stochastic differential equation models to produce smoother attribution maps, and later showed that shaping the injected noise further improves robustness (Jha et al., 2022). Walker et al. (2024) observed that gradients from saturation regions contribute noise and proposed Integrated Decision Gradients, which concentrates integration on the decision-relevant portion of the path. For vision transformers, metric-driven attributions (Walker et al., 2025b) and information-flow decompositions (Walker et al., 2025a) improve patch-level explanations. Guided IG (Kapishnikov et al., 2021) adapts the integration path to reduce pixel-level noise. Integration-budget sensitivity has also been studied directly. Makino et al. (2024) examine instance-dependent completeness residuals, nonmonotone changes with step count and unresolved path gradients. Singhi & Kumari (2024) investigate step-count sensitivity in IG variants and the Important Direction Gradient Integration (IDGI) framework; their reported stability experiment compares JPEG-compressed and uncompressed inputs. Our fixed-context diagnostics separately examine repeatability and coordinate changes across budgets. The smooth construction in Appendix B supplies a limitation on certification: agreement on finitely many grids and exact scalar completeness do not establish accurate coordinate integrals. Petsiuk et al. (2018) introduced the insertion and deletion curves we adopt as ranking-based faithfulness metrics. The reliability of perturbation metrics is itself an active concern.

Hooker et al. (2019) evaluate attributions by retraining on ablated inputs. Rong et al. (2022) show that deletion without retraining leaks information through the shape of the mask itself, and Tomsett et al. (2020) find that common saliency metrics are themselves unreliable. Our case study makes this distinction explicit for action-discrepancy responses: it relates their normalized curves algebraically and rescores preserved interventions while holding their feature ordering fixed. Metric dependence itself is already established by this literature. Adebayo et al. (2018) proposed model and data randomization tests to assess whether saliency maps depend on learned parameters and training data. Their results caution against interpreting visually plausible maps as evidence of that dependence. Our backbone randomizations are limited tests because learned encoders and adaptors remain in place. For diffusion generative models in the image domain, cross-attention maps relate generated pixels to prompt tokens (Tang et al., 2023). This differs from attributing a scalar discrepancy of a predicted control chunk.

Our implementation applies IG to a fixed-noise action-discrepancy function with modality-specific baselines. Path weighting is also an established attribution idea, as in IDG (Walker et al., 2024); the factor derived here follows specifically from changing the scalar discrepancy target. The analysis below specializes these established concerns to quadratic and stabilized negative-norm action discrepancies, tracing their separate effects on fixed interventions and integrated rankings.

2.2 Vision-Language-Action Models

VLA models unify perception, language understanding, and action generation. RT-2 (Zitkovich et al., 2023) fine-tunes a vision-language model to emit discretized actions as token sequences. Octo (Octo Model Team et al., 2024) trains a generalist policy on a large multi-robot dataset with a transformer backbone. Diffusion Policy (Chi et al., 2023) frames action generation as conditional denoising diffusion (Ho et al., 2020). It produces action chunks by iterative refinement rather than single-step regression. RDT (Liu et al., 2025) scales this approach with a Diffusion Transformer (DiT) backbone, a SigLIP vision encoder (Zhai et al., 2023), and a T5-XXL language encoder (Raffel et al., 2020). RDT processes six camera slots (three camera views over a two-frame history) through SigLIP into 4,374 vision tokens. It combines them with up to 1,024 language tokens and a 128-dimensional unified state vector, and it generates 64-step action chunks through a five-step DPM-Solver++ (Lu et al., 2025) denoising schedule. This multimodal, multi-step architecture makes RDT a compelling but challenging target for attribution.

2.3 Interpretability in Robotics

Prior work on explaining robotic policies has focused on reinforcement learning agents in discrete-action settings. Greydanus et al. (2018) applied perturbation-based saliency to Atari agents, and counterfactual methods (Amitai et al., 2024) contrast the outcomes of executed and alternative actions for policy explanation. In continuous control, VisualBackProp (Bojarski et al., 2018) visualizes convolutional neural network (CNN) saliency for end-to-end driving policies. That work addresses convolutional driving models rather than a multimodal diffusion sampler. Recent VLA studies examine different explanation targets. Zhang et al. (2026) estimate visual influence using randomized perturbations and action-discrepancy responses, with explicit assumptions for relating mean-squared error to distributional divergence. VLA-Trace (Shi et al., 2026) combines representation comparisons, attention interventions and behavioral probes. IMPACT-VLA (Kim & Park, 2026) attributes phase-modality interventions to downstream task success through counterfactual closed-loop execution. Our estimand is the discrepancy of a fixed-noise predicted action chunk at a fixed context, not an explanation of eventual task success. The present study examines how response geometry affects the evaluation of IG rankings for this estimand; it does not claim to originate VLA interpretability.

3 Attribution Targets and Evaluation

3.1 Integrated Gradients Preliminaries

Given a differentiable function $F: \mathbb{R}^n \rightarrow \mathbb{R}$, an input $x \in \mathbb{R}^n$, and a baseline $x' \in \mathbb{R}^n$, the integrated gradient along dimension i is

$$\text{IG}_i(x, x') = (x_i - x'_i) \int_0^1 \frac{\partial F}{\partial x_i} \Big|_{x' + \alpha(x - x')} d\alpha. \quad (1)$$

Under the fundamental-theorem and chain-rule assumptions along the path, IG satisfies *completeness*: $\sum_i \text{IG}_i(x, x') = F(x) - F(x')$. We approximate the integral with a uniform Riemann sum of m steps, giving

$$\text{IG}_i \approx \frac{x_i - x'_i}{m+1} \sum_{r=0}^m \frac{\partial F}{\partial x_i} \Big|_{x' + \frac{r}{m}(x - x')}. \quad (2)$$

This equal-weight endpoint average describes the historical implementation. Convergence requires regularity of the path integrand; finite-precision accumulation introduces additional error that increasing m does not isolate or necessarily reduce.

3.2 A Deterministic Target Through the Denoising Chain

At policy call t , write the sampler output as $g_\theta(o_t, z_t)$ for observation o_t and fixed diffusion noise z_t . We denote this predicted action chunk by μ_t for consistency with historical notation; it is a sample conditioned on fixed noise, not an identified conditional mean. The reference chunk $a_t = g_\theta(o_t, z_t)$ is computed once without gradients. For an intervened input, the primary attribution target is the auxiliary score

$$Q_t = -\frac{1}{2\sigma^2 D} \|a_t - \mu_t\|^2, \quad (3)$$

with $\sigma^2 = 1$ and $D = 512$ active entries: 64 predicted time steps over eight active dimensions. This is the nonconstant part of a Gaussian log kernel averaged over entries, with its log-normalizing constant omitted. It has not been derived as the diffusion policy’s conditional log likelihood. Historical code and files call this target `logpi`; we use ΔQ for its change. Attribution concerns the complete predicted chunk, of which the execution loop applies a subsampled prefix before replanning or termination.

We hold the noise realization fixed across the interpolation path to evaluate one deterministic scalar function. Attribution of an expectation over noise would be a different estimand and requires a justified gradient-expectation interchange and Monte Carlo treatment. Independent noise at each path point does not compute the fixed-noise quantity considered here. The historical implementation replaced nonfinite gradient entries by zero. That operation can corrupt individual coordinates while cancellation leaves the scalar completeness residual small or even zero; completeness is not a certificate of coordinate accuracy.

3.3 Alternative Scalar Targets

We compare the quadratic target with two alternative action-discrepancy targets,

$$y_t^{\ell_2} = -\sqrt{\|a_t - \mu_t\|_2^2 + \epsilon}, \quad y_t^{\max} = -\max_j |a_{t,j} - \mu_{t,j}|, \quad (4)$$

where $\epsilon = 10^{-12}$ is the implemented stabilizer and both targets use the same D active entries. The stabilized ℓ_2 target is smooth in the residual and has zero gradient at exact agreement; away from the stabilizer scale its action-space gradient approaches unit norm. The maximum-deviation target, abbreviated `maxdev`, uses a different norm and is not a monotone transform of ℓ_2 in general: residuals $(1, 1)$ and $(1.1, 0)$ reverse order. The quadratic and stabilized ℓ_2 responses are algebraically related for a fixed perturbation, although their gradients can induce different attribution rankings. Their units and scales differ, so a larger magnitude alone does not demonstrate greater information or operational usefulness. Section 5.1 isolates response changes for the quadratic and stabilized ℓ_2 targets. The maximum-deviation records remain an auxiliary diagnostic in the released analysis.

3.4 Response Transforms and Attribution Path Weights

Fix the reference action, active coordinates, model, noise and other modalities, and write $r(x) = \|a - \mu(x)\|_2^2$ and $L(x) = -\sqrt{r(x) + \epsilon}$. Where the gradients exist, the chain rule gives the exact identity

$$\nabla Q(x) = \frac{\sqrt{r(x) + \epsilon}}{D\sigma^2} \nabla L(x). \quad (5)$$

Thus nonzero pointwise gradients have the same direction. At exact self-reference both vanish. Along $x(\alpha) = x' + \alpha(x - x')$, however,

$$\text{IG}_i^Q = (x_i - x'_i) \int_0^1 \frac{\sqrt{r(x(\alpha)) + \epsilon}}{D\sigma^2} \partial_i L(x(\alpha)) d\alpha. \quad (6)$$

The positive weight generally varies with α . When coordinate contributions vary differently along the path, their integrated rankings can therefore change. This does not contradict IG implementation invariance, which compares implementations of the same scalar function (Sundararajan et al., 2017).

A concrete example uses the scalar action $\mu(x_1, x_2) = x_1^2 + \frac{4}{5}x_2$, baseline $(0, 0)$, actual input $(1, 1)$, detached reference $a = \frac{9}{5}$, and $D = \sigma^2 = 1$. On the same straight path, exact quadratic-target integration yields

$$\text{IG}^Q = \left(\frac{23}{30}, \frac{64}{75} \right), \quad \lim_{\epsilon \downarrow 0} \text{IG}^L = \left(1, \frac{4}{5} \right). \quad (7)$$

Quadratic IG ranks feature 2 first, whereas stabilized L with the implemented $\epsilon = 10^{-12}$ ranks feature 1 first with a margin at least 0.1999975. Appendix A proves the finite-stabilizer bound, so the reversal does not rely on replacing the implemented response by its limit. This establishes existence in a smooth nonlinear policy, not a universal rank change or an empirical RDT effect.

3.5 Per-Step Episode Attribution

RDT outputs action chunks of 64 steps, subsampled $4\times$ to 16 environment steps. We run IG once per *policy call*, not per environment step, because the chunk is the atomic unit of decision. The attribution unit is therefore one policy query. Runtime depends on the implementation, hardware, precision and integration budget; the earlier timing summaries are not used as a performance result.

At each policy call we compute three independent attributions through modality-specific forward functions.

- **Vision.** Freeze language and state and interpolate the adapted image tokens from the baseline to the real observation.
- **Language.** Freeze vision and state and interpolate the adapted language tokens from a minimal two-token baseline encoded through T5-XXL to the real instruction embedding.
- **State.** Freeze vision and language and interpolate the 128-dimensional unified state vector (eight active dimensions for the Franka arm, namely seven joints plus the gripper) from zeros to its real value. The state-adaptor multilayer perceptron (MLP) sits inside the forward function to preserve per-joint structure.

Algorithm 1 summarizes the procedure.

Baseline selection. The vision baseline is a uniform gray image in the external-camera slot only. The remaining five camera slots carry constant background images and are frozen at their real values, so IG attributes only to the informative slot. The language baseline is a minimal two-token sequence (the tokenizer’s padding and end-of-sequence tokens, since T5 has no beginning-of-sequence token) encoded through T5-XXL rather than a zero vector. The encoder output is zero-padded to the stored embedding length, and the language forward passes attend over the task’s real token span. The attended baseline is therefore the

Algorithm 1 Fixed-noise per-policy-call IG construction. Historical artifact availability is qualified in Section 4.

Input. Policy π_θ , environment $\mathcal{E}$, integration steps m , target $y \in \{Q, \ell_2, \text{maxdev}\}$

Output. Attributions $\{A_t^V, A_t^L, A_t^S\}$ at every policy call.

```

1:  $o_0 \leftarrow \mathcal{E}.\text{reset}()$ 
2: for policy call  $t = 0, 1, \dots$  do
3:   Encode  $o_t$  into vision  $v_t$ , language  $l_t$ , state  $s_t$ 
4:   Draw and store  $z_t$ ; compute  $a_t \leftarrow g_\theta(o_t, z_t)$  without gradients
5:   Define  $F$  from  $a_t$ , the fixed-noise predicted chunk  $\mu_t$ , and the chosen target
6:   for modality  $M \in \{V, L, S\}$  do
7:     Set baseline  $x'_M$ , input  $x_M$ 
8:     Freeze the other two modalities
9:      $A_t^M \leftarrow \text{IG}(F_M, x_M, x'_M, m)$ 
10:  end for
11:  Record  $A_t^V, A_t^L, A_t^S$  and completeness errors
12:  Apply action chunk and advance  $\mathcal{E}$  by up to 16 steps, stopping on termination
13: end for

```

Eq. 2

encoded token pair followed by zero vectors at the remaining attended positions. A zero vector is numerically permitted by the GELU-based language adaptor but defines a different representation baseline; preliminary efficacy comparisons are not independently reconstructible. The state baseline is a zero vector in the normalized joint space; its physical support and usefulness as an explanation reference are not established by this choice. Appendix C.2 reports the available vision-baseline sensitivity.

3.6 Evaluation Protocol

Completeness. For each modality, with the other inputs frozen, we report the absolute residual $|\sum_i \text{IG}_i - (F(x) - F(x'))|$ and its relative version

$$\text{Err} = \frac{|\sum_i \text{IG}_i - (F(x) - F(x'))|}{|F(x) - F(x')|}. \quad (8)$$

An exactly zero signed gap makes the relative quantity undefined. A small nonzero gap may amplify numerical error. The historical action-norm filter $\|a_t\| \geq 15$ does not bound this denominator, and is retained only as a population sensitivity. A low residual checks a scalar accounting identity; canceling coordinate errors can still change the attribution map or ranking.

Perturbation curves. We rank eligible feature groups by the sum of absolute coordinate attributions. The groups are external-camera patch tokens for vision and attended instruction tokens for language. Deletion replaces a prefix of the ranking with baseline features; insertion replaces a prefix of a baseline input with original features. For a scalar response R , the normalized ordinate is

$$\tilde{R}(x_k) = \frac{R(x_k) - R(x')}{R(x) - R(x')}. \quad (9)$$

We integrate on each record’s stored fraction grid. Neither the curve nor its AUC is necessarily bounded by $[0, 1]$. The historical grid requests $\{0, 1, 5, 10, 20, 30, 50, 75, 100\}\%$ of groups; rounding changes realized language-token fractions, so the saved curves cannot be treated as uniformly sampled continuous functions. Top- k score changes are $R(x_k) - R(x)$, with units defined by R . We distinguish the attribution target that generates a ranking from the response that evaluates its interventions. The earlier study-specific thresholds are not used as efficacy tests in the revised tables.

Randomization and baseline diagnostics. C1 reinitializes model parameters and correlates the resulting attributions with the originals. The standard variant changes the final action-output projection and

recomputes its reference action; frozen-reference and backbone-cascade variants retain the original reference. The historical cascade simultaneously reinitializes all backbone parameter matrices of rank at least two; it retains one-dimensional parameters, encoders and adaptors. It is not a progressive layerwise cascade. C2 shuffles both external-camera pixels and language. The historical implementation uses a pre-T5 shuffled-language cache when available and otherwise permutes post-T5 embeddings; the saved records do not authenticate which mode each call used. These are different dependence tests, and C2 is not the label-randomization experiment of Adebayo et al. (2018). Historical perturbed maps use $m = 16$ and original maps nominally use $m = 64$, so C1 also changes the numerical approximation. A baseline sensitivity compares black, gray and blurred observation features.

4 Retrospective Data and Statistical Protocol

Scope and traceability. The revised analysis authenticates 99 preserved files containing 22,966 valid records and 1 specifically identified corrupt line. Each valid physical record has a source path, original line number and byte hash. The raw inventory contains 129 repeated-key groups, of which 58 differ in scientific fields rather than only timing metadata. The last occurrence is a declared retrospective selection convention, not evidence that a conflicting attempt is correct. A separate sensitivity removes every key with a conflicting payload. The records, occurrence decisions, population membership, result identifiers and generated manuscript cells are linked in the released lineage registry. This authenticates preserved data bytes, not missing checkpoints, observations, action tensors or noise.

Campaigns and models. The principal reconstructed RDT-170M tables use the saved Month 7 verification campaign at $m = 64$, covering PickCube, StackCube, PegInsertionSide and PickSingleYCB in ManiSkill3 (Tao et al., 2024). These are separate experiments from the unavailable original main-pass inputs. The model was pretrained RDT-170M; the saved RDT-1B PickCube campaign used a task-finetuned checkpoint. The model comparison changes training, trajectories and historical preprocessing as well as parameter count, and is not a scale or competence intervention. Appendix C therefore reports the 1B cohort separately. No missing original completeness, standard-sanity, or early PickCube/StackCube integration-budget result is copied into a primary table.

Known protocol limits. The historical action chunk is subsampled before simulator execution, while attribution concerns the full predicted chunk. The four-task scope reflects historical implementation choices. An attempted TurnFaucet run used an incorrect controller, and no corrected rerun is documented; unsuccessful outcomes under that configuration cannot justify a capability exclusion. Early follow-up jobs used a zero language baseline in place of encoded padding/end tokens, and the PickSingleYCB follow-up used a proxy PickCube instruction embedding. The pooled 1B language cohort spans the early baseline and later encoded baselines. Early sidecar filenames also omitted seed and integration budget, permitting overwrites and possible reuse across nominal runs. Seed-tagged Month 7 records mitigate that filename issue but do not recover old contexts. We do not infer exact context equality from matching scalar fingerprints.

Estimands and uncertainty. The primary population is all valid saved calls after the declared occurrence rule. Continuous call-level summaries use medians over calls; residual pass rates use proportions over the stated call population. Solver contrasts report relative changes between endpoint medians, and the separately named equal-episode analysis uses episode medians. We resample entire recorded episode groups, keyed by task, model, recorded seed and episode, with $B = 10,000$ draws and seed zero. Point estimates and intervals use exactly the same source-record population. These percentile intervals describe the released empirical episode mixture, with longer episodes contributing more calls; they do not estimate variability across independently trained checkpoints. A separately named 1B sensitivity reports the median of episode medians, weighting episodes equally. For the old randomization variants we also group potentially reused nominal task/episode keys across seeds as a conditional dependence sensitivity. Neither grouping repairs unobserved context identity. Very small cluster counts are explicitly disclosed, and two-group baseline resampling is not presented as calibrated interval coverage. All unfiltered results and the historical norm-selected sensitivity remain available; no favorable tail is removed to obtain a threshold verdict.

The additional paired rescoring analysis defines $\Delta_{ej} = \text{AUC}_{Q,ej} - \text{AUC}_{L,ej}$ within call j of recorded episode e , holding that call's ranking and interventions fixed. Its primary estimand is $E^{-1} \sum_{e=1}^E n_e^{-1} \sum_{j=1}^{n_e} \Delta_{ej}$, with equal episode weights; the median of paired call differences is a prespecified descriptive sensitivity. These are distinct from the difference of the marginal medians above. The protocol specifies all eight ranking/modality/direction combinations before computing the additional summaries, with the same 10,000-draw episode-bootstrap convention. Their intervals are conditional, unadjusted descriptions; no calibrated joint coverage or global significance claim is made. A zero endpoint gap under either response is retained as undefined in the accounting. All 750 calls per ranking cohort have defined paired differences, with 25 calls in each of 30 recorded episodes, so equal-episode and equal-call means coincide here.

5 Response Geometry and Retrospective Results

5.1 Separating Attribution Target from Evaluation Response

The same recorded interventions can be rescored without recomputing IG or changing a selected feature. Let d be the recorded active-action discrepancy, $Q = -d^2/1024$, and $L = -\sqrt{d^2 + \epsilon}$ with $\epsilon = 10^{-12}$. Then

$$L = -\sqrt{-1024Q + \epsilon}, \quad Q = -\frac{L^2 - \epsilon}{1024}. \quad (10)$$

For an exact self-reference actual endpoint and an all-baseline squared residual $R > 0$, the normalized responses satisfy

$$\begin{aligned} N_Q(r) &= 1 - \frac{r}{R}, & N_L(r) &= \frac{\sqrt{R + \epsilon} - \sqrt{r + \epsilon}}{\sqrt{R + \epsilon} - \sqrt{\epsilon}}, \\ N_Q(r) - N_L(r) &= \frac{(\sqrt{r + \epsilon} - \sqrt{\epsilon})(\sqrt{R + \epsilon} - \sqrt{r + \epsilon})}{R}. \end{aligned} \quad (11)$$

Consequently $N_Q > N_L$ for $0 < r < R$, with equality at both endpoints and reversed ordering for $r > R$. Integration with nonnegative weights preserves the weak ordering if every intervention remains in range, even for a nonmonotone curve; strict area inequality requires positive integration weight on an interior residual. Overshooting points can change the area ordering and must not be clipped; if $R = 0$, both normalizations are undefined. For in-range curves, switching from Q to L therefore lowers insertion and deletion AUC together. If larger insertion AUC and smaller deletion AUC are interpreted as better ranking quality, this single response change suggests opposite quality changes for interventions that have not changed. The retrospective computation below transforms the recorded endpoints directly and does not assume exact historical self-reference. We transform every recorded curve ordinate, evaluate its endpoints under the transformed response, and reintegrate on the same saved grid. Each ranking cohort contains 750 calls from 30 episodes after duplicate handling. For quadratic-target vision rankings, median deletion AUC changes from 0.4479 under Q to 0.2908 under L . For stabilized- ℓ_2 rankings it changes from 0.2830 under L to 0.4404 under Q . Table 1 and Figure 1 include both modalities and both perturbation directions.

This establishes response-geometry sensitivity within each saved ranking cohort. It does not establish that Q-IG and L2-IG select the same features, that their rankings are equivalent, or that either is effective. The attempted cross-cohort join has different reference-action norms at every nominal key; exact raw contexts and noise are unavailable. Apparent agreement after common rescoring therefore cannot replace a prospective paired ranking comparison. Likewise, the larger native L2 score is not by itself recovered information, and stabilized L2 has no square-root kink at exact agreement.

Paired changes and overshooting tails. Table 2 shows why a central summary alone is insufficient. All eight paired-call medians are positive, ranging from 0.1041 to 0.1590, whereas six equal-episode means are negative. For Q-ranking vision insertion, the median paired change is 0.1042 but the mean is -17.0966 , with conditional interval $[-33.8210, -3.9746]$. The most negative paired change in that case is -4438.8386 ; these finite extremes are retained without clipping or trimming. An exploratory influence check, reported for all eight cases in Appendix D, finds that all four negative insertion means remain negative after any one recorded episode is omitted. The two negative language-deletion means can change sign; their conditional

Table 1: Within-ranking response transformation, with median AUC and conditional 95% episode-bootstrap intervals. Each ranking cohort contains 750 calls from 30 episodes. The same recorded interventions are reused within a ranking cohort. The two ranking cohorts are not authenticated paired contexts; comparisons between them do not identify ranking improvement.

Ranking target	Response	Modality	Insertion AUC	Deletion AUC
Q	Q	Vision	0.883 [0.845, 0.909]	0.448 [0.408, 0.504]
Q	Q	Language	0.580 [0.510, 0.658]	0.372 [0.311, 0.428]
Q	L	Vision	0.756 [0.718, 0.785]	0.291 [0.253, 0.333]
Q	L	Language	0.469 [0.412, 0.516]	0.257 [0.228, 0.284]
L	Q	Vision	0.884 [0.833, 0.921]	0.440 [0.402, 0.501]
L	Q	Language	0.635 [0.582, 0.720]	0.368 [0.302, 0.407]
L	L	Vision	0.756 [0.700, 0.803]	0.283 [0.252, 0.327]
L	L	Language	0.507 [0.456, 0.566]	0.249 [0.202, 0.278]

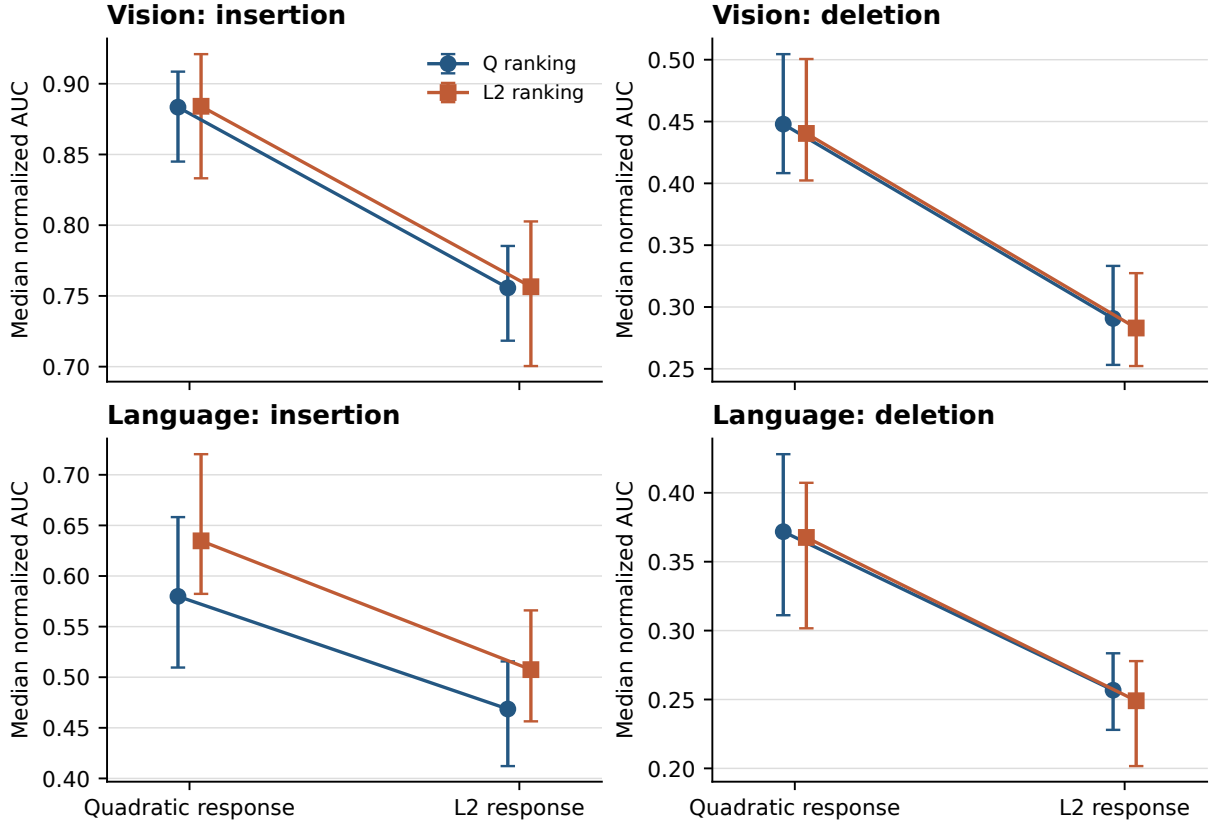


Figure 1: Changing only the evaluation response while keeping each recorded ranking and its interventions fixed. Error bars are conditional episode-bootstrap intervals for each median. Lines connect responses within a ranking cohort; the two ranking cohorts are not authenticated pairs. There is no universal chance reference line.

intervals already cross zero. This diagnoses concentration in the recorded cohort, not population robustness, and does not change the untrimmed primary estimands.

Table 2: Paired response differences $\Delta = \text{AUC}_Q - \text{AUC}_L$ within the same saved curve. Each row uses 750 calls from 30 recorded episodes; rows within a ranking cohort reuse those calls. The primary mean weights episodes equally. Its 95% interval is a conditional, unadjusted episode-bootstrap description. The paired-call median is shown alongside it because overshooting tails can reverse the sign of the mean. Neither summary is a comparison of Q and L2 ranking quality.

Ranking	Modality	Curve	Mean Δ	95% interval	Median Δ
Q	vision	insertion	-17.0966	[-33.8210, -3.9746]	0.1042
Q	vision	deletion	0.1227	[0.1059, 0.1382]	0.1588
Q	lang	insertion	-0.1819	[-0.4051, -0.0277]	0.1169
Q	lang	deletion	-0.0095	[-0.1022, 0.0631]	0.1189
L2	vision	insertion	-18.5620	[-37.8672, -3.1731]	0.1041
L2	vision	deletion	0.0184	[-0.2037, 0.1378]	0.1590
L2	lang	insertion	-0.0329	[-0.0963, 0.0266]	0.1279
L2	lang	deletion	-1.3710	[-4.2081, 0.0780]	0.1205

We classify each saved squared-discrepancy ordinate relative to the interval between its stored actual and baseline endpoint discrepancies, without assuming exact self-reference. High-distance overshoots exceed both endpoints; low-distance undershoots fall below both. The artifact records endpoint orientation and retains both kinds of departure, undefined gaps and a prespecified boundary-label sensitivity. In Q-ranking vision insertion, 312 of 750 calls contain at least one high-distance overshoot, and 89 have a negative paired AUC change. Decomposing the trapezoidal difference by saved-node category gives mean contributions of 0.1144 from in-range nodes and -17.2111 from high-distance overshoots, with no low-distance contribution in that case. These add to the reported mean and locate the sign reversal in the recorded out-of-range tail. The L2-ranking vision-insertion cohort shows the same qualitative mean/median disagreement in its own records. Node categories and segment endpoint labels describe only sampled interventions; they do not establish unobserved continuous curve behavior. The result qualifies the in-range ordering in Equation 11: response changes need not move a whole empirical curve population in one direction.

5.2 An Analytic Random-Order Counterexample

Consider the finite additive scalar policy $f(x) = \sum_{i=1}^n x_i$, an all-one input, a zero baseline and the self-reference $f(x) = n$. Every feature ordering gives the same perturbation values. Whole-feature prefix interventions realize only fractions $q = k/n$, with $k = 0, \dots, n$; at these fractions, the normalized quadratic deletion and insertion values are $1 - q^2$ and $1 - (1 - q)^2$. Trapezoidal integration over all $n + 1$ prefix fractions gives

$$\text{AUC}_n = \frac{2}{3} - \frac{1}{6n^2}$$

for either curve. Integrating their polynomial extensions over continuous $q \in [0, 1]$ instead gives $2/3$, also the limit as $n \rightarrow \infty$; it is not the exact finite-grid area. With $n = 100$, every fraction on the historical nine-point requested grid is realized exactly, and trapezoidal integration gives 0.65976 for both curves and every ordering. This finite example suffices to refute a universal random-order AUC of 0.5. An unstabilized negative-norm response instead has linear values on the same fractions and trapezoidal AUC of 0.5. Consequently, a fixed AUC reference such as 0.5 requires justification for the policy, response and intervention protocol; it is not universal. This construction is an analytic control, not an empirical random-ranking measurement on RDT.

5.3 Completeness and Perturbation Summaries

Table 3 reports completeness residuals from the saved verification campaign. Vision and language have smaller median relative residuals than state in these cohorts, but this does not establish more accurate

coordinates or a causal explanation based on dimension count. The historical integrator combines model-forward, path, summation and quadrature errors. Actual-model validation must separate those factors on identical inputs before selecting production settings.

Table 3: Month 7 verification campaign, all valid calls, $m = 64$. Cells are median relative completeness residuals in percent with conditional 95% episode-bootstrap intervals. Calls and episodes are the identical point/interval populations. These are historical numerical diagnostics, not coordinate-accuracy certifications.

Task	Calls	Episodes	Vision (%)	Language (%)	State (%)
PegInsertion	700	100	2.17 [1.94, 2.35]	1.83 [1.69, 1.98]	3.48 [3.00, 4.04]
PickCube	397	100	2.20 [1.90, 2.43]	1.88 [1.62, 2.30]	5.88 [4.99, 7.29]
PickYCB	397	100	1.85 [1.71, 2.06]	2.02 [1.79, 2.18]	4.26 [3.50, 5.62]
StackCube	400	100	2.39 [2.18, 2.59]	1.76 [1.58, 1.97]	10.53 [8.07, 13.13]

Table 4 reports native quadratic-response insertion and deletion AUCs for the same verification campaign. These are descriptive perturbation results. Without matched random and simple ranking controls, their magnitudes do not establish attribution efficacy. Normalization can produce extreme negative values: the smallest saved PickCube vision insertion AUC in the all-valid population is -18190.36. The analysis retains that observation and its endpoint denominator instead of clipping it or treating the action-norm filter as a validity certificate. Medians are consequently accompanied by the released denominator/tail diagnostics; a central summary alone does not describe the entire curve population.

Table 4: Month 7 verification campaign, all valid calls and the native quadratic response. Medians and conditional 95% intervals share one population. No entry is labeled a success against an uncalibrated chance threshold. The recorded token-fraction grids, endpoint conventions and provenance limits remain part of these descriptive estimates.

Task	Modality	Calls	Episodes	Insertion AUC	Deletion AUC
PegInsertion	Vision	700	100	0.731 [0.688, 0.774]	0.280 [0.234, 0.309]
PegInsertion	Language	700	100	0.320 [0.257, 0.379]	0.341 [0.291, 0.382]
PickCube	Vision	397	100	0.822 [0.771, 0.838]	0.167 [0.137, 0.213]
PickCube	Language	397	100	0.525 [0.468, 0.580]	0.433 [0.391, 0.475]
PickYCB	Vision	397	100	0.844 [0.808, 0.863]	0.337 [0.296, 0.372]
PickYCB	Language	397	100	0.335 [0.227, 0.409]	0.471 [0.438, 0.500]
StackCube	Vision	400	100	0.821 [0.777, 0.849]	0.208 [0.161, 0.261]
StackCube	Language	400	100	0.359 [0.292, 0.432]	0.480 [0.429, 0.512]

6 Numerical Diagnostics from an Incomplete Campaign

The new diagnostic campaign concerns identified current computations, separately from the historical records above. Historical rescoring evaluates arithmetic on saved curves whose original contexts cannot be recovered. The later probe evaluates a separately identified offline fp32 function on newly authenticated contexts, with its own reference actions. It neither validates the historical attribution maps nor supplies a paired comparison with them. It was stopped for compute cost before its planned numerical coverage and independent verification were complete. The analysis uses a fixed, authenticated snapshot of sealed results; unfinished entries are not classified as successful or failed calculations. This stopping decision provides no statistical sampling rule for the completed subset, so its criterion counts are descriptive rather than estimates of population failure rates.

Functions and precision controls. Authenticated stored observations, masks, reference actions and exact noise arrays permit comparisons on fixed inputs. The diagnostic implementation separates accumulation arithmetic, master-path and final-product arithmetic, quadrature and downstream model precision.

It compares full coordinate arrays, eligible-group ranks, selected groups and perturbation responses. An earlier downstream-fp32 factor control lifts the same bf16-rounded parameters and cached adapted representations, explicitly converts non-parameter dtype attributes, and holds the original reference action fixed. That comparison changes downstream arithmetic while holding the numeric inputs fixed; it does not recover unrounded checkpoint parameters or test fp32 encoders. Its fixed original reference also distinguishes it from the subsequent probe. The earlier comparisons preceded the full repeated-gradient checks, so they do not alone establish a globally repeatable precision contrast.

The later engineering campaign uses a separately specified deterministic fp32 function: image and language representations retain their cached bf16 values lifted to fp32, raw state is re-adapted in fp32, and the function generates its own fp32 action reference. It evaluates contexts visited by the bf16 policy with the stored source noise. This is an offline intervention function, not the bf16 behavior policy or a derivative of its executed commands. Repeating its endpoints tests a different property from repeating its path gradients and attribution maps. Changing the function or its reference cannot be treated as a quadrature-only comparison.

Engineering design and interpretation. The predeclared engineering bank spans four RDT-170M tasks and two tasks with an independently identified authors’ 1B checkpoint. Its identity does not recover or authenticate the checkpoint used by the historical task-finetuned 1B cohort in Appendix C. Source collection stopped after at most 13 policy calls, subject also to a 400-environment-step cap and earlier termination; call indices 0 and 12 were selected in advance. These are fixed-index engineering contexts, not uniformly sampled calls from complete trajectories. Collection did not select episodes by success or action norm. The integration ladders use trapezoidal quadrature with fp32 path and accumulation arithmetic. The same gradient stream supplies a mean path-gradient map without the IG input-difference multiplier; it is included in the numerical checks, not evaluated as an efficacy baseline. The numerical protocol compares coordinates, ranks, completeness and common perturbation responses across budgets and repeated calculations. Its tolerance checks diagnose the tested approximation; they are not efficacy margins or calibrated bounds on the error against an exact attribution. The finest available map is a comparator, not ground truth. The preserved cohort analysis reports every planned context, including calls not reached by the source rollout and entries without a sealed result.

Preserved coverage. The retained snapshot is labeled October 1, 2026, 03:36 UTC, before the later compute stop. Table 5 retains all 24 planned contexts and 144 target/modality cells. Twelve contexts, representing six episodes from the four 170M tasks, have all six diagnostics complete, giving 72 complete cells. Three planned calls were not reached by their source rollouts, accounting for 18 cells. The remaining nine available contexts, or 54 cells, lack sealed diagnostics at this cutoff. There is no completed 1B numerical context in this snapshot. These are snapshot statuses, not assertions about the final state of later interrupted work.

Table 5: Full planned engineering roster at the authenticated 03:36 UTC snapshot. The first four numerical columns count contexts; the last gives complete/planned target-modality cells. Each context has six cells. Unreached source calls and available but unsealed results remain distinct. Completion does not imply numerical qualification.

Stratum	Planned	Complete	Unreached	Unsealed	Cells
PegInsertionSide 170M	4	2	0	2	12/24
PickCube 170M	4	4	0	0	24/24
PickCube 1B	4	0	2	2	0/24
PickSingleYCB 170M	4	2	0	2	12/24
StackCube 170M	4	4	0	0	24/24
StackCube 1B	4	0	1	3	0/24
Total	24	12	3	9	72/144

Repeatability and budget sensitivity. The image and language ladders use $m = 128, 256, 512$; the state ladder uses $m = 256, 512, 1024$. Finest maps were repeated twice, and seven fixed path locations were probed three times each. All recorded repeated-condition equality and coverage checks satisfied their conditions. They are dependent checks on the same finite contexts, not independent replications or evidence of a population reliability rate.

Table 6 reports every completed target/modality group using the two finest recorded budgets. For maps A_m and A_{2m} , the coordinate discrepancy is $\|A_m - A_{2m}\|_1 / \|A_{2m}\|_1$, undefined at a zero denominator. The predeclared relative-coordinate and completeness thresholds are each 1%; additional checks concern group ordering and common-response sensitivity. The table’s violation counts refer to context/target/modality arms with at least one pairwise-threshold violation in this finer-budget comparison, including the mean path-gradient diagnostics; finest-budget completeness is summarized separately. All lower-budget comparisons, raw denominators and individual check results remain in the artifact. The maxima summarize the recorded cases without an interval or population generalization.

Table 6: Diagnostics on all 12 completed contexts per target/modality arm. n counts contexts, and “Viol.” counts arms with at least one recorded pairwise-threshold violation between the two finest budgets. The remaining columns are maxima across those contexts: relative IG coordinate L_1 difference, finest-budget relative completeness residual, and maximum raw RMS response-curve change divided by the all-baseline RMS displacement, all in percent. RMS maxima include IG and mean-path-gradient rankings and both perturbation directions. Coordinate differences use the finer map’s L_1 norm. Comparators are 256 versus 512 for vision/language and 512 versus 1024 for state. These dependent engineering diagnostics are not population error rates or proof that the finest maps are accurate; no setting or subset is approved.

Modality	Target	n	Viol.	IG L_1 (%)	Residual (%)	RMS (%)
Language	L	12	0	0.402	0.315	0.000
Language	Q	12	0	0.006	0.002	0.000
State	L	12	2	20.778	6.655	18.573
State	Q	12	2	32.119	11.436	1.641
Vision	L	12	4	1.234	0.288	7.361
Vision	Q	12	1	2.049	0.124	0.000

Both targets retain finest-budget state completeness violations at the two recorded PickSingleYCB episode-1 contexts, alongside substantial changes between the two finest maps. Their scalar gaps exceed the declared near-zero cutoffs; the corresponding signed gaps and absolute residuals are retained in the artifact. These discrepancies therefore cannot be dismissed as division by an almost-zero scalar gap. A state case also has unchanged intervention response curves despite substantial coordinate change, showing why response stability alone is insufficient to qualify the integral. Since the quadratic target also fails, the failures cannot all be assigned to the stabilized negative-norm target’s endpoint behavior. The sparse stored path probes do not identify an exclusive numerical cause.

Repeated equality can coexist with integration-budget sensitivity. A deterministic calculation can reproduce an approximation whose value changes with the integration budget, while a small completeness residual can conceal coordinate errors through cancellation. Accordingly, neither repeatability nor a single completeness statistic qualifies the production setting. Appendix B shows why even agreement across several budgets is insufficient in general: equal and opposite coordinate errors can preserve exact completeness while all selected coarse grids give the same ranking and response curves, with an ordering different from exact IG. Correctness here means recovery of the specified IG integral; perturbation-ranking quality remains a separate question. This analytic possibility does not identify the cause of the observed RDT failures. The observed failures support a bounded numerical diagnostic conclusion; the incomplete campaign does not establish accuracy across modalities, tasks, checkpoints or future contexts.

7 Discussion and Limitations

The reconstructible evidence separates three questions: what a scalar response measures, whether a numerical attribution is reliable, and whether its ranking contains useful model-dependent information. The saved response transformation answers the first question for fixed recorded interventions. It cannot answer the latter two by itself. The present claims concern exact relationships and reconstructed measurements, supported by the explicitly bounded numerical diagnostics. A new paired study with common responses and informative controls would be needed to establish ranking efficacy; no such efficacy conclusion is drawn here. The elementary identities are not presented as a new general theory of attribution. Their value here is to identify which operation changes a reported evaluation and to prevent that change from being misread as empirical ranking improvement. A common response itself defines the comparison’s estimand, rather than a universal measure of explanation quality. Its action coordinates, units, baselines, perturbation groups and aggregation require justification for the intended use. For the unresolved ranking question, matched random, input-difference and path-gradient controls would help distinguish information in the attribution from simple input structure and the IG multiplier. No comparative or learned-weight-specific result from the deferred protocols is claimed in this paper.

A practical evaluation procedure. The results suggest four steps when reporting action-discrepancy attributions.

1. Specify the scalar attribution target and evaluation response separately, together with the reference action, fixed noise, feature representation, baseline, intervention groups and schedule.
2. Compare rankings on authenticated matched contexts under each common response. Include matched random and simple ranking controls when claiming efficacy. Analyze a response change separately by rescoreing the same interventions, recomputing the endpoint normalization and retaining overshoots and undefined gaps.
3. Check repeatability and coordinate, ranking and response sensitivity across numerical settings on a declared context roster, retaining failures and incomplete cases. Completeness and agreement on finitely many grids are diagnostics, not certificates of exact IG.
4. Report the unit of analysis and dependence grouping, and distinguish conclusions about response sensitivity, numerical reliability and ranking quality. None of these alone establishes improved closed-loop behavior.

This procedure states what comparisons identify; it does not supply a universally adequate integration budget or evaluation response.

Historical uncertainty also has several distinct sources. Source-record hashing can authenticate released bytes while checkpoint and context identities remain unknown. Episode resampling addresses within-episode dependence conditional on a particular empirical mixture; it does not repair shared or overwritten contexts, preprocessing confounds, or missing training replicates. The 1B results come from one checkpoint, and the solver and alternative-target analyses cover narrower task populations than the verification campaign. Normalized response curves may be unbounded and sensitive to endpoint gaps. Camera-token and language-token interventions differ in group counts and semantics, so their AUCs are not directly interchangeable measures of modality importance.

Historical overlay panels are preserved as artifacts but are not used as evidence of successful grounding or behavioral failure diagnosis. Likewise, groups previously named from completeness residuals are numerical diagnostic groups, not independently validated labels such as misgrounding or delayed credit. The unrecovered original main-pass and early budget-extension numbers are excluded from revised primary tables. Appendix F reports only reconstructible later budget cohorts, without a convergence claim.

8 Conclusion

Per-policy-call IG for a fixed-noise diffusion sampler can be defined using an explicit auxiliary action-discrepancy score. For saved RDT perturbation curves, changing only the evaluation response changes normalized AUC without changing the selected features. Paired changes have positive medians in all eight reported cases, but overshooting tails make six means negative, showing why both the summary statistic and the endpoint geometry matter. A finite-grid analytic example rules out a universal random-ranking AUC of 0.5. The canonical retrospective tables preserve what the available records support and exclude unrecovered primary constants. The actual-model diagnostics retain unresolved numerical failures and incomplete coverage. The smooth aliasing construction further shows that finite-grid agreement can coexist with an ordering different from exact IG. A common response is necessary for an interpretable comparison of ranking quality, but our results do not establish that one target or attribution method is superior under it.

Ethical Statement

This work analyzes pretrained robotic manipulation policies in simulation; no human subjects were involved. Attribution is a diagnostic tool here, not a safety guarantee for robot deployment. Misinterpreting attractive maps, low completeness residuals or uncalibrated AUCs could create unjustified confidence in a policy, which motivates the explicit limits and controls.

Reproducibility Statement

The retrospective analysis parses a hash-authenticated raw inventory, records occurrence decisions, and links each result to physical source records. The revised tables and figures are generated from the same canonical artifact, with identical point/interval populations and a manuscript lineage registry. It retains conflicting-key exclusions, action-norm strata, endpoint/tail diagnostics, alternative episode weightings and conditional grouping sensitivities. The separate paired rescoring artifact provides its pre-computation analysis protocol, all eight cases, paired call and episode summaries, bootstrap membership, and an exact decomposition by saved-node geometry. A separately versioned exploratory supplement records every single-episode omission and its source identities without changing those primary results. The smooth nested-grid example includes an analytic proof and repeated FP32/FP64 CPU checks through the unchanged trapezoidal kernel. These operations reproduce arithmetic over saved records; they do not recover missing models, sidecars or original observations. The portable numerical projection retains the full planned roster, recorded checks and metrics, report hashes, and selected YCB FP32 tensors with their tensor hashes; it does not include all raw artifacts for the 72 completed target-modality cells. Its CPU reproducer recomputes the reported diagnostics and saved-tensor checks from this projection, not the complete original raw-artifact audit or a new model execution. The new runtime separately records immutable context, source, checkpoint, language, environment and noise identities, and fails explicitly on nonfinite attribution rather than sanitizing coordinates. A candidate PDF or a successful script run alone does not constitute an authenticated scientific replication. Anonymous manuscript sources and a research supplement accompany this audit copy. The identified repository URL is omitted to support independent review.

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A Response-Geometry Derivations

The identities in Section 3.4 assume that the relevant path gradients are integrable and hold the action reference fixed. Differentiating $Q = -r/(2D\sigma^2)$ and $L = -\sqrt{r+\epsilon}$ gives Equation 5 immediately. For Equation 11, set $a_0 = \sqrt{\epsilon}$, $b = \sqrt{R+\epsilon}$ and $z = \sqrt{r+\epsilon}$. Since $R = b^2 - a_0^2$ and $r = z^2 - a_0^2$, rationalization gives

$$\frac{b^2 - z^2}{R} - \frac{b - z}{b - a_0} = \frac{(z - a_0)(b - z)}{R}. \quad (12)$$

For in-range residuals the maximum difference is $(b - a_0)^2/(4R) \leq 1/4$. These pointwise statements concern a fixed ranking's interventions; they neither order different rankings nor establish task-success improvements.

For the rank-reversal example, let $c = 4/5$ and parameterize the path by (t, t) . The scalar residual is $d(t) = 1 + c - t^2 - ct = (1 - t)(1 + t + c) \geq 0$. The quadratic path integrands are $2td(t)$ and $cd(t)$, yielding

$$\text{IG}^Q = \left(\frac{1}{2} + \frac{c}{3}, \frac{2c}{3} + \frac{c^2}{2} \right) = \left(\frac{23}{30}, \frac{64}{75} \right). \quad (13)$$

For the stabilized negative norm the respective integrands are $2td(t)/\sqrt{d(t)^2 + \epsilon}$ and $cd(t)/\sqrt{d(t)^2 + \epsilon}$. Using $d(t) \geq c(1 - t)$ and monotonicity of $1 - d/\sqrt{d^2 + \epsilon}$ gives

$$\int_0^1 \left(1 - \frac{d(t)}{\sqrt{d(t)^2 + \epsilon}} \right) dt \leq 1 - \frac{\sqrt{c^2 + \epsilon} - \sqrt{\epsilon}}{c} \leq \frac{\sqrt{\epsilon}}{c}. \quad (14)$$

Therefore $0 \leq 1 - \text{IG}_1^L \leq 2\sqrt{\epsilon}/c = 2.5 \times 10^{-6}$ and $0 \leq c - \text{IG}_2^L \leq \sqrt{\epsilon} = 10^{-6}$ at the implemented stabilizer. All coordinates are positive and the stabilized L margin is at least 0.1999975, while Q has the opposite ordering with exact margin 13/150. The artifact includes independent differentiation, integration and normalization checks for these identities, including overshoots, zero gaps and a policy example without rank reversal.

B A Smooth Counterexample to Nested-Grid Agreement

Agreement between finitely many integration grids, even with exact completeness and unchanged perturbation curves, does not by itself certify the coordinate integrals. The following construction isolates this limitation without a model experiment. All equalities in the construction are statements in real arithmetic.

For a positive integer M , define the smooth scalar policy and target

$$h(t) = \frac{2}{5} \sin^2(\pi Mt), \quad (15)$$

$$\mu(x, y) = \frac{2}{5}x + \frac{3}{5}y + (x - y)h\left(\frac{x + y}{2}\right), \quad (16)$$

$$Q(x, y) = -\frac{1}{2}(\mu(x, y) - 1)^2. \quad (17)$$

Take baseline $(0, 0)$, actual input $(1, 1)$ and the fixed own-reference action $\mu(1, 1) = 1$. This has the same quadratic self-reference form as the main target, with one active action entry and $\sigma^2 = 1$. Along the straight path (t, t) , the transverse term in μ vanishes, so $\mu(t, t) = t$, $Q(t, t) = -(1 - t)^2/2$, and

$$\nabla Q(t, t) = (1 - t) \left(\frac{2}{5} + h(t), \frac{3}{5} - h(t) \right). \quad (18)$$

The endpoint score gap is $1/2$. Both coordinate integrals are positive and their exact values are distinct.

Let T_m denote componentwise composite trapezoidal integration with m intervals. Whenever m divides M , every node $t = k/m$ has $h(t) = 0$. The remaining coordinate functions are linear, hence

$$T_m = \left(\frac{1}{5}, \frac{3}{10} \right), \quad \mathbf{1}^\top T_m = \frac{1}{2} = Q(1, 1) - Q(0, 0). \quad (19)$$

Thus all such grids have zero completeness residual and the same strict absolute-coordinate ranking, with feature 2 first. They also generate identical insertion and deletion masks under a fixed deterministic ranking rule. Evaluating those masks on the same policy gives identical response curves for every such grid, regardless of whether its coordinate integrals are accurate.

In contrast, $h(1 - t) = h(t)$ and $\int_0^1 h(t) dt = 1/5$, giving

$$\int_0^1 (1 - t)h(t) dt = \frac{1}{10}, \quad \text{IG}(Q) = \left(\frac{3}{10}, \frac{1}{5} \right). \quad (20)$$

The exact ranking puts feature 1 first. The coarse-grid errors are $(-1/10, 1/10)$: their sum cancels exactly, while their coordinate L_1 norm is $1/5$, or 40% of the exact map’s L_1 norm. The path score itself is a quadratic polynomial; unresolved variation lies in its coordinate-gradient decomposition, which a scalar completeness check cannot recover.

For this particular construction, the $2M$ grid resolves the discrepancy: its sampled h values alternate between 0 and $2/5$, so $T_{2M}h = 1/5$. Grid symmetry gives $T_{2M}[(1 - t)h] = \frac{1}{2}T_{2M}h = 1/10$, recovering Equation 20. This is not a universal sufficient doubling rule. For any fixed finite list of uniform interval counts, choosing M as a common multiple places all of those grids in Equation 19. The example supplies no derivative bound that would otherwise control the unresolved variation.

Exact integration and perturbation quality are different questions. For the displayed $M = 128$, $h(1/2) = 0$, so the whole-feature interventions have $\mu(1, 0) = 2/5$ and $\mu(0, 1) = 3/5$. With the same normalized Q response and grid $\{0, 1/2, 1\}$, the aliased feature-2-first ordering has insertion ordinates $(0, 0.84, 1)$ and deletion ordinates $(1, 0.64, 0)$, giving AUCs 0.67 and 0.57. The exact-IG feature-1-first ordering instead has insertion AUC 0.57 and deletion AUC 0.67. Thus the inaccurate map’s ranking scores better under both conventional preferences, larger insertion AUC and smaller deletion AUC, in this toy. The finite off-path interventions measure a different property from the diagonal path integral: the transverse policy term changes its coordinate gradients without changing these Boolean intervention values. Neither completeness nor exact integration guarantees an optimal perturbation score. This is a toy-specific separation, not evidence of RDT ranking efficacy.

The standalone artifact `analysis/revision/nested_grid_aliasing.py` evaluates the unchanged production trapezoidal IG kernel on this CPU toy with $M = 128$, in FP32 and FP64. Budgets 32, 64, 128 reproduce the incorrect order, whereas 256 reproduces the analytic order to numerical tolerance. The report retains measured coordinate errors, completeness residuals, repeated-run equality and common response curves; it does not replace finite-precision results with the exact equalities above. The two-coordinate example uses no RDT weights or observations. It explains why finite-grid agreement alone is insufficient for certification, not why any particular PickSingleYCB diagnostic failed, and approves no production integration setting.

C Secondary Historical Results

These analyses preserve additional observations from the saved campaigns. Their training, reference, context and preprocessing qualifications prevent interpreting them as model-scale effects, learned-weight-specific efficacy or a causal test of denoiser contraction.

C.1 The Separately Recorded 1B Cohort

The saved 1B analysis illustrates a second distinction: changing the episode weighting changes the estimand. Table 7 uses the same 1,199 calls from 60 episodes under both summaries. The language summaries are more sensitive to this weighting than the vision summaries, within a cohort that also mixes language-baseline provenance. Neither estimate is a training-replication result or a controlled improvement over 170M.

Table 7: Saved RDT-1B PickCube campaign, 1,199 source calls and 60 episodes under both estimands. The first weights each call equally; the second weights each episode’s median equally. Conditional 95% episode intervals describe the released mixture. Evaluation seeds share a checkpoint, and the pooled language population spans two baseline provenances.

Estimand	Modality	Insertion AUC	Deletion AUC
Median over calls	Vision	0.940 [0.926, 0.954]	0.288 [0.250, 0.312]
Median over calls	Language	0.541 [0.449, 0.605]	0.267 [0.187, 0.327]
Median of episode medians	Vision	0.940 [0.927, 0.959]	0.289 [0.235, 0.313]
Median of episode medians	Language	0.488 [0.368, 0.581]	0.297 [0.151, 0.342]

C.2 Randomization and Baseline Diagnostics

Table 8 reconstructs the standard verification-campaign correlations. Model randomization retains substantial rank association, while input shuffling changes the maps more strongly. That contrast does not make model dependence unnecessary or certify semantic grounding. A map based only on input magnitude can change under input shuffling while ignoring learned policy weights. The saved frozen-reference and cascade variants, including a dependence-grouping sensitivity, are reported in Appendix E.

Table 8: Month 7 model randomization (C1) and input shuffle (C2), all recorded calls. Conditional 95% episode intervals accompany median Spearman correlation. These diagnostics test different dependencies: low C2 correlation does not establish learned-model dependence, and C1 changes integration budget as well as weights.

Task	Check	Calls	Episodes	Vision ρ	Language ρ
PegInsertion	C1	100	16	0.960 [0.949, 0.969]	0.959 [0.947, 0.966]
PegInsertion	C2	100	16	0.089 [0.067, 0.111]	0.018 [0.004, 0.036]
PickCube	C1	100	26	0.956 [0.948, 0.961]	0.926 [0.917, 0.944]
PickCube	C2	100	26	0.142 [0.128, 0.157]	0.117 [0.099, 0.150]
PickYCB	C1	100	26	0.943 [0.930, 0.949]	0.917 [0.903, 0.931]
PickYCB	C2	100	26	0.164 [0.149, 0.175]	0.128 [0.081, 0.155]
StackCube	C1	100	26	0.958 [0.952, 0.965]	0.965 [0.960, 0.969]
StackCube	C2	100	26	0.140 [0.117, 0.154]	0.266 [0.246, 0.279]

The IG multiplier $(x_i - x'_i)$ permits shared structure but does not force a particular ranking. The path gradients can dominate the result, and the input features themselves come from retained learned encoders/adaptors. Persistent correlations therefore motivate input-only and path-gradient-only controls; they do not establish multiplier dominance. Spearman correlation is not a fraction of model-independent information. The baseline comparisons in Appendix E show imperfect agreement but cannot establish equivalence with their small number of groups.

C.3 Action Displacement and Solver Resolution

The quadratic score is a normalized squared displacement, so its numerical magnitude has a direct dependence on the chosen convention. Removing $1/D$ multiplies exact scores and attributions by D without changing exact ranks, normalized AUC, or relative completeness. This algebra does not determine whether an action change is operationally important; that requires action units and a justified behavioral criterion.

The saved solver analysis compares median relative action displacement under top-5% vision deletion between $T = 2$ and $T = 20$. Figure 2 reports the endpoint contrast with conditional resampling of nominal episode pairs. Reference actions vary across solver settings, and exact contexts/noise cannot be authenticated. The intervals permit appreciable increases and decreases. Changing T changes discretization over the full

denoising horizon, rather than adding steps along one fixed trajectory. These observations therefore identify neither a causal contraction mechanism nor its absence.

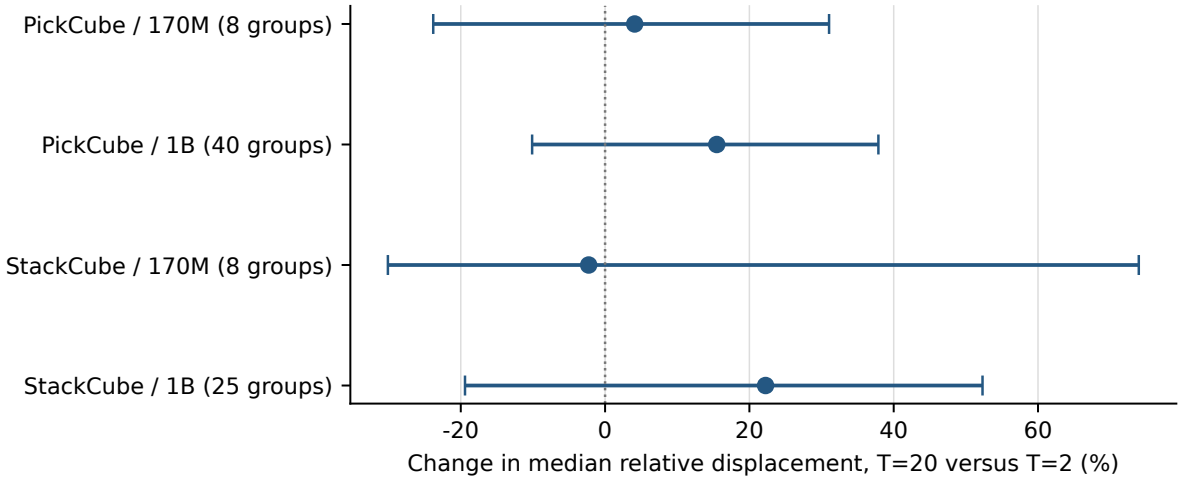


Figure 2: Retrospective change in median relative displacement under top-5% vision deletion. Intervals use nominal episode pairing across solver resolutions, conditional on the released records and grouping. The vertical reference denotes no endpoint change, not a contraction test or an equivalence margin. Raw context/noise identity remains unknown.

D Exploratory Episode Influence

The paired summaries in Table 2 are sensitive to large finite response changes. To locate that sensitivity, we omit each of the 30 recorded episodes in turn, retain all calls in the other 29 episodes, and recompute their equally weighted episode mean. Table 9 reports every case. This analysis followed an exploratory audit of the same records; its separately saved protocol is not an outcome-blind preregistration. The omission ranges are deterministic influence summaries, not confidence intervals or evidence that the recorded episodes form an independent sample. The full untrimmed means, medians and conditional intervals remain unchanged.

Table 9: Exploratory omissions of one complete recorded episode. The range spans all 30 omission means; “Same sign” counts agreement with the full-cohort mean. Each case uses 750 calls in 30 episodes before omission; rows in a ranking cohort reuse calls. These ranges are not confidence intervals. All 240 omission records and their retained-population hashes are supplied in the separate influence artifact.

Ranking	Modality	Curve	Full mean	Omission-mean range	Same sign
Q	vision	insertion	-17.0966	[-17.6910, -11.5667]	30/30
Q	vision	deletion	0.1227	[0.1208, 0.1272]	30/30
Q	lang	insertion	-0.1819	[-0.1937, -0.0932]	30/30
Q	lang	deletion	-0.0095	[-0.0158, 0.0182]	27/30
L2	vision	insertion	-18.5620	[-19.2071, -11.9674]	30/30
L2	vision	deletion	0.0184	[0.0128, 0.1264]	30/30
L2	lang	insertion	-0.0329	[-0.0399, -0.0193]	30/30
L2	lang	deletion	-1.3710	[-1.4246, 0.0328]	29/30

All four negative insertion means remain negative under every individual omission. The Q language-deletion mean changes sign in 3 of 30 omissions and the L2 language-deletion mean in 1 of 30; both already have conditional bootstrap intervals spanning zero. In the latter case the most negative episode contributes 97.29% of the sum of magnitudes of negative episode means. This is a concentration measure, not a share of the signed net mean and not a reason to exclude that episode.

The extreme vision-insertion changes also have a concrete normalization explanation. For the most negative Q-ranking call, the baseline squared residual is 0.0174195, the smallest in its 750-call cohort, whereas the maximum saved perturbed residual is 653.053. Their ratio is about 37,490. The finite Q endpoint gap is 1.7011×10^{-5} , above the historical 10^{-9} cutoff; the native Q AUC is -4464.9082 and its L2 rescore is -26.0696 . A small baseline discrepancy and a large intervention response therefore coexist. These values are not an undefined or machine-zero division and do not provide an action-unit calibration of physical importance. The artifact gives the same diagnostics within each separate ranking cohort; matching nominal episode labels do not authenticate a cross-arm pair.

E Additional Retrospective Diagnostics

The frozen-reference and backbone-cascade analyses retain learned encoders/adaptors and historical context uncertainty. Table 10 reports both recorded-episode grouping and a conditional shared-context grouping. These are sensitivity analyses, not proof that either grouping is independent. Table 11 reports baseline-ranking point summaries; the small group count prevents an equivalence conclusion.

Table 10: Saved backbone-randomization variants. Shared-context grouping joins nominal task/episode keys across recorded seeds as a dependence sensitivity; it does not authenticate shared inputs. Intervals are conditional on the stated grouping. Learned encoders/adaptors remain, and sidecar reuse and language-baseline deviations remain unresolved.

Variant	Grouping	Calls	Groups	Vision ρ	Language ρ
Frozen	Recorded episodes	400	16	0.929 [0.905, 0.940]	0.862 [0.835, 0.879]
Frozen	Shared-context sensitivity	400	8	0.929 [0.904, 0.947]	0.862 [0.820, 0.886]
Cascade	Recorded episodes	400	16	0.713 [0.696, 0.738]	0.910 [0.896, 0.920]
Cascade	Shared-context sensitivity	400	8	0.713 [0.693, 0.763]	0.910 [0.894, 0.926]

Table 11: Saved baseline-ranking correlations. Only two nominal episode groups are available per task. Interval output is retained in the analysis artifact as a descriptive resampling diagnostic, but is not presented here as calibrated uncertainty. Positive correlation does not establish baseline interchangeability.

Task	Calls	Groups	Black/gray	Black/blur	Gray/blur
PickCube	30	2	0.825	0.539	0.455
StackCube	30	2	0.814	0.546	0.496

F Saved Integration-Budget Cohorts

Table 12 replaces the previous table of unrecovered and duplicated constants with the reconstructible later cohorts. The norm-selected population and occurrence rule are retained explicitly for this historical sensitivity. These calls are not paired with the earlier $m = 64$ main pass on identical observations, baselines, runtime and noise. Truncating trajectories to the same call depth does not create such pairs. An integration-budget effect must instead be evaluated on authenticated fixed contexts, with precision and quadrature controlled.

Table 12: Reconstructible $m = 128$ cohorts only. These conditional intervals describe recorded cohorts, not convergence under paired contexts. The final row removes every conflicting repeated key. Missing original Pick-Cube/StackCube subsets are not reconstructed or replaced with constants.

Task	Seed	Calls	Groups	Vision residual $\leq 3\%$ (%)
PegInsertion	42	189	8	64.0 [50.3, 76.9]
PegInsertion	142	160	7	76.2 [67.1, 85.0]
PickYCB	42	194	8	73.7 [62.1, 85.5]
PickYCB	142	195	8	71.8 [60.9, 83.1]
PegInsertion (conflicts excluded)	142	142	7	75.4 [63.8, 85.0]